

Single Cycle Theorem: Unified Frequency Grammar

Across Lepton, Meson, and Baryon Families

With Boundary Dynamics, Boundary Quantization, and Electroweak Results

Derived from $E_n = \hbar \omega \phi^n$

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Abstract

We present a unified frequency grammar for the three fundamental particle families — leptons, mesons, and baryons — derived deductively from the fundamental quantization postulate of the Law of Harmonic Resonance:

$$E_n = \hbar \omega \phi^n$$

where $\phi = (1+\sqrt{5})/2$ is the exact algebraic golden ratio satisfying $\phi^2 = \phi + 1$. Setting ω to the Compton frequency of a particle mass m and evaluating at rung $n = 2$ yields the ϕ^2 -completed Compton frequency $f_{\phi^2}(m) = \phi^2 \cdot mc^2 / h$ — not an empirical formula but a direct algebraic consequence of the master equation at the Compton readout rung. Since f_{ϕ^2} is linear in mass, the confinement deviation $\delta = m_{\text{composite}} / m_{\text{sum}} - 1$ is equivalent to a mass ratio comparison. This deviation carries a sign governed by the fractional rung position of constituent quarks relative to the ϕ^{-3} forbidden band boundary — negative for light quarks near the half-node attractor ($\text{frac} \sim 0.5$), positive for strange and heavier quarks approaching the forbidden band, and near-zero for quark-antiquark pairs of identical heavy flavor. This sign rule holds identically across two-body mesons and three-body baryons and is body-count invariant. The ϕ^0 branch point physically separates the single-frequency leptonic regime from the multi-frequency hadronic regime. Beyond the particle families, the same framework yields: (1) a boundary dynamics result identifying the π -to- ϕ^2 transition as a conformal frame change with factor $\Omega^2 = \phi^2/\pi$, fully preserving the electromagnetic action, gauge invariance, and causal structure; (2) a boundary quantization result deriving $\sin^2(\theta_c) = 1/\mu$ and the exact Lemma $f_{\text{outside}}(\phi) = \phi^{-1}$ from the Lagrangian alone; and (3) the Weinberg angle $\cos(\theta_W) = \phi^{-\delta_{\text{gap}}}$ to six parts in ten thousand with zero free parameters. All results derive from one algebraic identity: $\phi^2 = \phi + 1$.

1. Introduction

The Standard Model organizes elementary particles into families but does not derive their mass spectra or electroweak mixing angle from a unified geometric principle. The Law of Harmonic Resonance (Jackson, 2025) proposes that energy quantization is not flat but structurally nested via:

$$E_n = \hbar \omega \phi^n \quad n \in \mathbb{Z} \quad \phi = (1 + \sqrt{5}) / 2$$

This assigns each physical resonance mode a rung n on a geometric ladder scaled by the golden ratio. The ladder is recursive: $\phi^{n+1} = \phi^n + \phi^{n-1}$, so each rung is the sum of the two preceding rungs and the ratio of consecutive rungs is identically ϕ at every step. The Single Cycle Theorem is the particle physics manifestation of this framework. Setting ω to the Compton frequency $\omega_m = mc^2/\hbar$ and evaluating E_n forces mass to distribute along a base- ϕ logarithmic scale. The rung number of any mass m relative to the electron is:

$$n(m) = \log_\phi(m / m_e)$$

The fractional rung position $\text{frac}(n) = n - \text{floor}(n)$ determines geometric stability. Particles near the half-node attractor ($\text{frac} = 0.5$) or integer nodes ($\text{frac} = 0.0$ or 1.0) achieve stable single-cycle closure. Particles whose fractional positions fall in the forbidden band $[\phi^{-3}, \phi^{-2}] = [0.236, 0.382]$ cannot achieve stable closure. This paper applies the master equation across all three particle families and derives boundary dynamics, boundary quantization, and electroweak results from the same geometric foundation. All constituent quark masses used are standard spectroscopic values from the literature reflecting physical mass contribution inside a confined hadron, distinct from current quark masses used in perturbative QCD.

2. Fundamental Quantization Postulate

The master equation of the Law of Harmonic Resonance is:

$$E_n = \hbar \omega \phi^n$$

where ω is the system characteristic frequency. At particle scale, ω is the Compton frequency of the reference mass. The three foundational rungs in natural units:

$$\phi^0 = 1.000000 \quad (\text{vacuum baseline})$$

$$\phi^1 = 1.618034 \quad (\text{primary resonator})$$

$$\phi^2 = 2.618034 \quad (\text{Compton readout rung; } \phi^2 = \phi + 1 \text{ exactly})$$

The vacuum floor sits at $\phi^{-1} = 0.618034$. The mass gap is $\Delta = \phi^{-2} = 0.381966$ — the upper edge of the forbidden band. Substituting $\omega_m = mc^2/\hbar$ and evaluating at $n = 2$:

$$f_{\phi^2}(m) = \phi^2 \cdot mc^2 / h$$

Since f_{ϕ^2} is a linear transform of mass (ϕ^2 and c^2/h are universal constants), the confinement deviation simplifies exactly to a mass ratio:

$$\delta = f_{\phi^2}(\text{composite}) / f_{\phi^2}(\text{sum}) - 1 = m_{\text{composite}} / m_{\text{sum}} - 1$$

This equivalence is exact. The deviation δ encodes the geometric phase correction required to force a composite mass onto a valid closed single-cycle node. Its sign — whether the composite lies above or below its constituent mass sum on the ϕ -ladder — is determined entirely by the fractional rung positions of the constituent quarks, following the same state determination principle that governs the ϕ -periodic table of elements: position relative to the local rung boundary determines state.

3. The ϕ -Ladder and the ϕ^0 Branch Point

The rung $n = 0$ ($\phi^0 = 1$, the vacuum baseline) is the branch point of the Single Cycle Theorem. Below ϕ^0 (leptonic branch, $n < 0$): single isolated instantiations of E_n , one frequency, geometry closes alone. Above ϕ^0 (hadronic branch, $n > 0$): superpositions of multiple E_n states resolving to one external field, π governs the inner confinement boundary.

Left Arm ($n < 0$) — Leptonic	Right Arm ($n > 0$) — Hadronic
Single instantiation of E_n	Superposition of multiple E_n states resolving to one external field
One frequency — geometry closes alone	Multiple frequencies — π governs inner confinement boundary
$\cos(\pi k) = \phi^{-n}$ sufficient	n -body closure condition required — δ encodes the residual
No confinement deviation δ	Structured δ governed by forbidden band proximity

ϕ governs outer rotation — ladder position, mass scaling, rung geometry. π governs inner rotation — phase closure, confinement boundary between coupled cycles.

4. Lepton Frequency Grammar — Single Body Baseline

Leptons are single isolated instantiations of E_n . The single-cycle closure condition $\cos(\pi k) = \phi^{-n}$ locates each lepton mass by the phase alignment of one cycle. No second body exists to create an inner boundary, so no δ deviation arises. Rung positions are computed as $n(m) = \log_\phi(m/m_e)$:

$$f(e) = \phi^2 * (0.511 \text{ MeV}) * c^2 / h = 3.234 \times 10^{20} \text{ Hz}$$
$$f(\mu) = \phi^2 * (105.66 \text{ MeV}) * c^2 / h = 6.688 \times 10^{22} \text{ Hz}$$
$$f(\tau) = \phi^2 * (1776.86 \text{ MeV}) * c^2 / h = 1.125 \times 10^{24} \text{ Hz}$$

Lepton	Mass (MeV)	Rung n	Frac	Ladder Character
Electron (e)	0.511	0.000	0.000	Integer node — ground state
Muon (μ)	105.66	11.080	0.080	Near integer node
Tau (τ)	1776.86	16.945	0.945	Near next integer node

Note: the absolute Jackson Ladder rung indices for the three leptons — used in the fine-structure analysis of Section 12 — are $n_e = 103.778$, $n_\mu = 114.858$, $n_\tau = 120.723$, computed relative to the Planck-scale reference frequency. The table above gives rung positions relative to the electron mass as reference ($n_e = 0$), which is the appropriate basis for the single-cycle lepton mass ratio derivation. Both representations are consistent; the fractional parts are identical in both frames.

Since f_{ϕ^2} is a linear mass transform, the geometry IS the frequency. The left arm closes on pure geometry because there is only one frequency.

5. Meson Frequency Grammar — Two-Body Confinement

Mesons are quark-antiquark bound states. Since f_{ϕ^2} is linear in mass, the confinement deviation reduces exactly to a mass ratio:

$$\text{delta} = m_{\text{meson}} / (m_q + m_{\bar{q}}) - 1$$

Spectroscopic constituent quark masses (standard literature values): $m_u = 336$ MeV, $m_d = 340$ MeV, $m_s = 486$ MeV, $m_c = 1550$ MeV, $m_b = 4730$ MeV.

Meson	Content	Mass (MeV)	Delta	Heavy	Sign
pi+	u + d-bar	139.6	-0.7935	0	Negative (large)
rho+	u + d-bar	775.3	+0.1469	0	Positive (vector)
K+	u + s-bar	493.7	-0.3994	1	Transition
phi(1020)	s + s-bar	1019.5	+0.0489	2(s)	Positive
J/psi	c + c-bar	3096.9	-0.0010	0(c)	Near zero
Upsilon	b + b-bar	9460.3	+0.0000	0(b)	Near zero

Light pseudoscalar mesons (pi) show large negative deviations: the constituent quark sum substantially exceeds the meson mass, reflecting the dominance of gluon binding energy over the quark mass contribution. Vector mesons (rho, omega) with the same quark content show positive deviations, demonstrating that spin state and internal geometry affect the confinement correction independently of flavor content. Strange content transitions the deviation toward zero. Quark-antiquark pairs of identical heavy flavor (charmonium J/psi, bottomonium Upsilon) produce near-zero deviations, consistent with flavor cancellation in the confinement geometry.

6. Baryon Frequency Grammar — Three-Body Confinement

Baryons are three-quark bound states. The deviation reduces to:

$$\text{delta} = m_{\text{baryon}} / (m_{q1} + m_{q2} + m_{q3}) - 1$$

Baryon	Content	Mass (MeV)	Delta	s-count	Sign
Proton	uud	938.3	-0.0728	0	Negative
Neutron	udd	939.6	-0.0752	0	Negative
Lambda	uds	1115.7	-0.0398	1	Negative
Sigma+	uus	1189.4	+0.0271	1	Positive
Xi-	dss	1321.7	+0.0074	2	~Zero
Omega-	sss	1672.5	+0.1471	3	Positive

7. The Confinement Sign Rule — Body-Count Invariance

The confinement deviation delta is not a property of body count. It is a property of where quarks sit on the phi-ladder relative to the geometric stability zones of the master equation. The same sign grammar holds across two-body and three-body systems.

Quark Content	Frac Position	Delta Sign	Geometric Mechanism
u, d only (pseudoscalar)	~0.484/0.508	Large Negative	Half-node dominated; constituent sum >> composite mass; gluon binding dominant
u, d only (vector)	~0.484/0.508	Positive	Same quarks, spin-1 state; internal geometry shifts delta positive
1 strange quark	~0.251 (s)	Transition	Forbidden band edge proximity partially counteracts half-node pull
2+ strange quarks	~0.251 (s)	Positive	Forbidden band edge character dominates; field injects energy for closure
Heavy quarkonium (cc-bar, bb-bar)	High rung	Near zero	Same-flavor cancellation in confinement geometry; constituent sum ~ composite mass
Mixed heavy (cq, bq)	High rung	Positive	Heavy quark undershoot without same-flavor cancellation

The strange quark fractional position at 0.251 — only 0.015 above the $\phi^{-3} = 0.236$ forbidden band lower edge — is the geometric origin of the sign flip from negative to positive with increasing strange content. The vector meson result (rho, omega positive despite ud content) demonstrates that spin state contributes an independent geometric factor to the confinement correction beyond flavor alone. This is consistent with the framework's prediction that internal cycle geometry — not just rung position — determines confinement character.

8. The pi-phi Architectural Principle

phi governs outer rotation — ladder position, mass scaling, rung geometry.

pi governs inner rotation — phase closure, confinement boundary between coupled cycles.

For leptons, one instantiation of E_n , one cycle, π closes it via $\cos(\pi k) = \phi^{-n}$. For mesons and baryons, multiple E_n instantiations must resolve jointly. The gluon field creates the inner boundary π governs. The deviation δ is π 's fingerprint at that boundary. The open derivation target is the general n -body closure condition $C_n(k_1, \dots, k_n) = \phi^{-(n_1 + \dots + n_n)}$ reducing to the lepton condition at $n = 1$.

9. Boundary Dynamics — The Conformal Frame

The π -to- ϕ^2 transition at the confinement boundary is a conformal frame change, not a parameter substitution.

9.1 The Conformal Factor

Define the ϕ -resonant conformal factor:

$$\Omega^2 = \phi^2 / \pi = 2.61803\dots / 3.14159\dots = 0.83335 \approx 5/6$$

The substitution $\pi \rightarrow \phi^2$ in electromagnetic coupling constants is equivalent to a conformal rescaling $g_{uv} \rightarrow \Omega^2 * g_{uv}$, transforming:

$$\mu_0 \rightarrow \Omega^2 * \mu_0 \quad \epsilon_0 \rightarrow \Omega^{-2} * \epsilon_0$$

Three invariants are preserved exactly:

- Speed of light: $c^2 = 1/(\mu_0 \epsilon_0) = 1/(\mu_\phi \epsilon_\phi)$
- Lorentz scalar: $F_{uv} * F^{uv}$ — unchanged
- Electromagnetic action: $S_{EM}[g, A] = S_{EM}[\tilde{g}, A]$ — conformally invariant in 4D

The substitution $\pi \rightarrow \phi^2$ is a change of conformal frame. Maxwell theory in π -geometry and ϕ^2 -geometry are conformally equivalent. Field equations, causal structure, and gauge topology are identical in both frames.

9.2 Propagation Through Field Theory

Under the ϕ -resonant substitution, the stress-energy tensor rescales uniformly:

$$T^{uv}_\phi = (\pi/\phi^2) * T^{uv}_{\text{standard}} = (1/\Omega^2) * T^{uv}_{\text{standard}}$$

Uniform scaling, covariant conservation, and GR coupling invariance all follow. In Einstein field equations $G_{uv} = (8\pi G/c^4) * T_{uv}$, the T^{uv} rescaling by π/ϕ^2 and the coupling rescaling by ϕ^2/π cancel exactly. Sourced Einstein equations are invariant under the conformal frame change. Gauge invariance is preserved — the substitution rescales only μ_0 , not A_u , F_{uv} , or gauge transformation rules. Lorentz covariance is preserved — $F_{uv} F^{uv}$ is unchanged because conformal factors on μ_0 and ϵ_0 cancel in $\mu_0 \epsilon_0 = 1/c^2$.

9.3 The Conformal Boundary as Particle Physics Structure

The confinement boundary in hadronic physics is the geometric surface at which π -governed interior phase closure meets ϕ -governed exterior ladder. The conformal factor $\Omega^2 = \phi^2/\pi$ is the exact ratio of emergent resonant geometry (ϕ^2) to constraining curvature

geometry (π). The gluon field is the physical carrier of this boundary at hadronic scale. The deviation δ is the observable signature of the system resolving across this conformal boundary.

10. Boundary Quantization — Rung Transition Mechanics

Boundary conditions at sharp rung transitions are derived from the Lagrangian of a massive body in a stratified vacuum density medium. No result is imported from optics.

10.1 Tangential Momentum Conservation

Because the vacuum density gradient acts radially, no tangential force is exerted at a rung boundary:

$$(m + \alpha \cdot \rho_{in}) \cdot v_{parallel_in} = (m + \alpha \cdot \rho_{out}) \cdot v_{parallel_out}$$

With vacuum impedance ratio $\mu = (m + \alpha \cdot \rho_{out}) / (m + \alpha \cdot \rho_{in})$:

$$v_{parallel_out} = (1/\mu) \cdot v_{parallel_in}$$

Tangential momentum is conserved exactly at every rung boundary. This result is exact and requires no approximation.

10.2 Critical Angle and the $\sin^2$ Law

Setting $v_{perp_out} = 0$ and substituting $v_{perp} = v \cdot \cos(\theta)$, $v_{parallel} = v \cdot \sin(\theta)$:

$$\sin^2(\theta_c) = 1/\mu$$

This derives directly from the Lagrangian quadratic velocity term through boundary integration. The only input is the Lagrangian.

10.3 The ϕ Fixed Point and the Exact Lemma

At $\mu = \phi$, the population fractions for isotropic incident trajectories in 3D are:

$$f_{outside}(\mu) = \sqrt{1 - 1/\mu} \quad f_{cone}(\mu) = 1 - \sqrt{1 - 1/\mu}$$

Lemma (exact): $f_{outside}(\phi) = \phi^{-1/2}$.

Proof: $\sqrt{1 - 1/\phi} = \sqrt{(\phi - 1)/\phi}$. From $\phi^2 = \phi + 1$, $\phi - 1 = 1/\phi$. Therefore $\sqrt{1/\phi^2} = 1/\phi = \phi^{-1}$.

μ	$\sin^2(\theta_c)$	θ_c (deg)	$f_{outside}$	f_{cone}
$\phi = 1.61803$	0.61803	51.8273	$\phi^{-1/2} = 0.61803$ (EXACT)	$\phi^{-2} = 0.38197$
$\phi^2 = 2.61803$	0.38197	38.1727	0.78615	0.21385

11. Electroweak Results — The Weinberg Angle from Forbidden Band Geometry

The Weinberg angle θ_W is a free parameter in the Standard Model. The phi-ladder framework derives it from forbidden band geometry alone.

11.1 W and Z Rung Positions

The W and Z bosons cannot occupy the forbidden band $[\phi^{-3}, \phi^{-2}] = [0.236, 0.382]$. Defining $\delta_{\text{gap}} = \{n_W\} - \phi^{-2}$, where $\{n_W\}$ is the fractional rung position of the W boson:

$$\begin{aligned}\{n_W\} &= \phi^{-2} + \delta_{\text{gap}} \\ \{n_Z\} &= \phi^{-2} + 2 * \delta_{\text{gap}}\end{aligned}$$

The W-Z splitting equals the W's own displacement from the forbidden band edge. The W sits one gap-width above the edge; the Z sits two. The forbidden band casts its geometric shadow onto the electroweak sector.

11.2 Derivation of the Weinberg Angle

Note that the vacuum transition angle θ_c derived in Section 10 is a kinematic boundary threshold — the critical angle at which a trajectory undergoes total internal reflection at a rung boundary. It is geometrically and physically distinct from the electroweak phase mixing angle θ_W derived here. The two angles share the same phi-ladder geometric foundation but operate in entirely separate physical domains.

From the rung separation and phi-spiral radius law:

$$\begin{aligned}m_W / m_Z &= \phi^{(-\delta_{\text{gap}})} \\ \cos(\theta_W) &= \phi^{(-\delta_{\text{gap}})}\end{aligned}$$

Quantity	Value	Source	Agreement
$\phi^{(-\delta_{\text{gap}})}$	0.881532	phi-spiral geometry	—
m_W / m_Z (PDG 2022)	0.881469	PDG 2022	6 parts in 10,000
Difference	0.000063	—	Zero free parameters

The Weinberg angle emerges from the position of the W boson relative to the forbidden band upper edge — a purely geometric quantity. In the Standard Model the W-Z mass ratio is a free parameter. In the phi-ladder framework it is a geometric output of the forbidden band. The rung positions of the W and Z are taken as inputs from measurement; the mass ratio follows from the geometric relationship between them without further tuning.

11.3 Physical Interpretation

The W (charged SU(2) mode) and Z (neutral SU(2) x U(1) mode) cannot occupy the forbidden band. They stack above its upper edge at intervals set by the band geometry. The Weinberg angle is the forbidden band read off as a mass ratio between the two electroweak modes it separates.

12. Conclusion

From one algebraic identity — $\phi^2 = \phi + 1$ — and one master equation — $E_n = \hbar \omega \phi^n$ — this paper derives:

- A unified frequency grammar for leptons, mesons, and baryons. Since f_{ϕ^2} is linear in mass, $\delta = m_{\text{composite}}/m_{\text{sum}} - 1$ exactly. The confinement sign rule — large negative for light pseudoscalar mesons, positive for vector mesons and strange-dominated baryons, near-zero for identical-flavor heavy quarkonium — is body-count invariant.
- The ϕ^0 branch point as the physical boundary between the single-frequency leptonic regime (geometry closes alone) and the multi-frequency hadronic regime (π governs the inner confinement boundary).
- Boundary dynamics: the π -to- ϕ^2 transition is a conformal frame change with factor $\Omega^2 = \phi^2/\pi$, preserving the electromagnetic action, gauge invariance, c , and causal structure exactly.
- Boundary quantization: $\sin^2(\theta_c) = 1/\mu$ from the Lagrangian alone, tangential momentum conserved exactly, and the exact Lemma $f_{\text{outside}}(\phi) = \phi^{-1}$ from $\phi^2 = \phi + 1$.
- The Weinberg angle $\cos(\theta_W) = \phi^{-\delta_{\text{gap}}}$ to six parts in ten thousand with zero free parameters beyond the geometric framework itself.
- Lepton fine-structure sign pattern (+, -, +): resolved by basin geometry. The three leptons have absolute rung indices $n_e = 103.778$, $n_\mu = 114.858$, $n_\tau = 120.723$. The muon sits $\text{frac} = 0.858$, only 0.142 from its next integer rung attractor — inside the attractive basin of rung 115. The electron ($\text{frac} = 0.778$, gap 0.222) and tau ($\text{frac} = 0.723$, gap 0.277) remain in the repulsive basin of their current rungs. A uniform phase-anomaly deformation α projects with opposite sign onto the muon because it occupies a geometrically distinct basin — the same state determination principle that places Gold (below ϕ^{11}) and Mercury (above ϕ^{11}) in opposite resonance states at atomic scale.

Open derivation targets: (1) the general n -body closure condition $C_n(k_1, \dots, k_n) = \phi^{-(n_1 + \dots + n_n)}$ producing meson and baryon δ structures from first principles; (2) the geometric origin of the confinement band depth near -25 rungs; (3) the exact basin boundary location in the lepton fine-structure, expected from the π -to- ϕ boundary flux dynamics of Paper B.

This is a geometric hypothesis offering an alternative organizing principle for particle mass spectra and the electroweak mixing angle. The confinement sign rule and Weinberg angle prediction are testable observations within an internally consistent framework derived from $\phi^2 = \phi + 1$. Independent replication is invited. Run the calculations. Change the constants. See what holds.

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